In [26]:
```
pip install opencv-python matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: opencv-python in c:\users\asus\appdata\roaming\python
\python312\site-packages (4.13.0.92)
Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib\site-packa
ges (3.9.2)
Collecting numpy>=2 (from opencv-python)
  Using cached numpy-2.4.4-cp312-cp312-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-pac
kages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-
packages (from matplotlib) (24.1)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packag
es (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anaconda3\lib
\site-packages (from matplotlib) (2.9.0.post0)
INFO: pip is looking at multiple versions of contourpy to determine which version is
compatible with other requirements. This could take a while.
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-package
s (from python-dateutil>=2.7->matplotlib) (1.16.0)
Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
Using cached numpy-2.4.4-cp312-cp312-win_amd64.whl (12.3 MB)
Installing collected packages: numpy, contourpy
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.4
    Uninstalling numpy-1.26.4:
      Successfully uninstalled numpy-1.26.4
Successfully installed contourpy-1.3.3 numpy-2.4.4
Note: you may need to restart the kernel to use updated packages.
  WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\Users\ASUS
\AppData\Roaming\Python\Python312\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning,
use --no-warn-script-location.
ERROR: pip's dependency resolver does not currently take into account all the packag
es that are installed. This behaviour is the source of the following dependency conf
licts.
gensim 4.3.3 requires numpy<2.0,>=1.18.5, but you have numpy 2.4.4 which is incompat
ible.
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.4.4 which is incompatib
le.
scipy 1.13.1 requires numpy<2.3,>=1.22.4, but you have numpy 2.4.4 which is incompat
ible.
```

In [27]:
```
import cv2
import matplotlib.pyplot as plt
```

In [28]:
```
img = cv2.imread('image.jpg')

if img is None:
    print("Error: Image not found. Check file name/path.")
else:
    print("Image loaded successfully")
```

```
Image loaded successfully
```

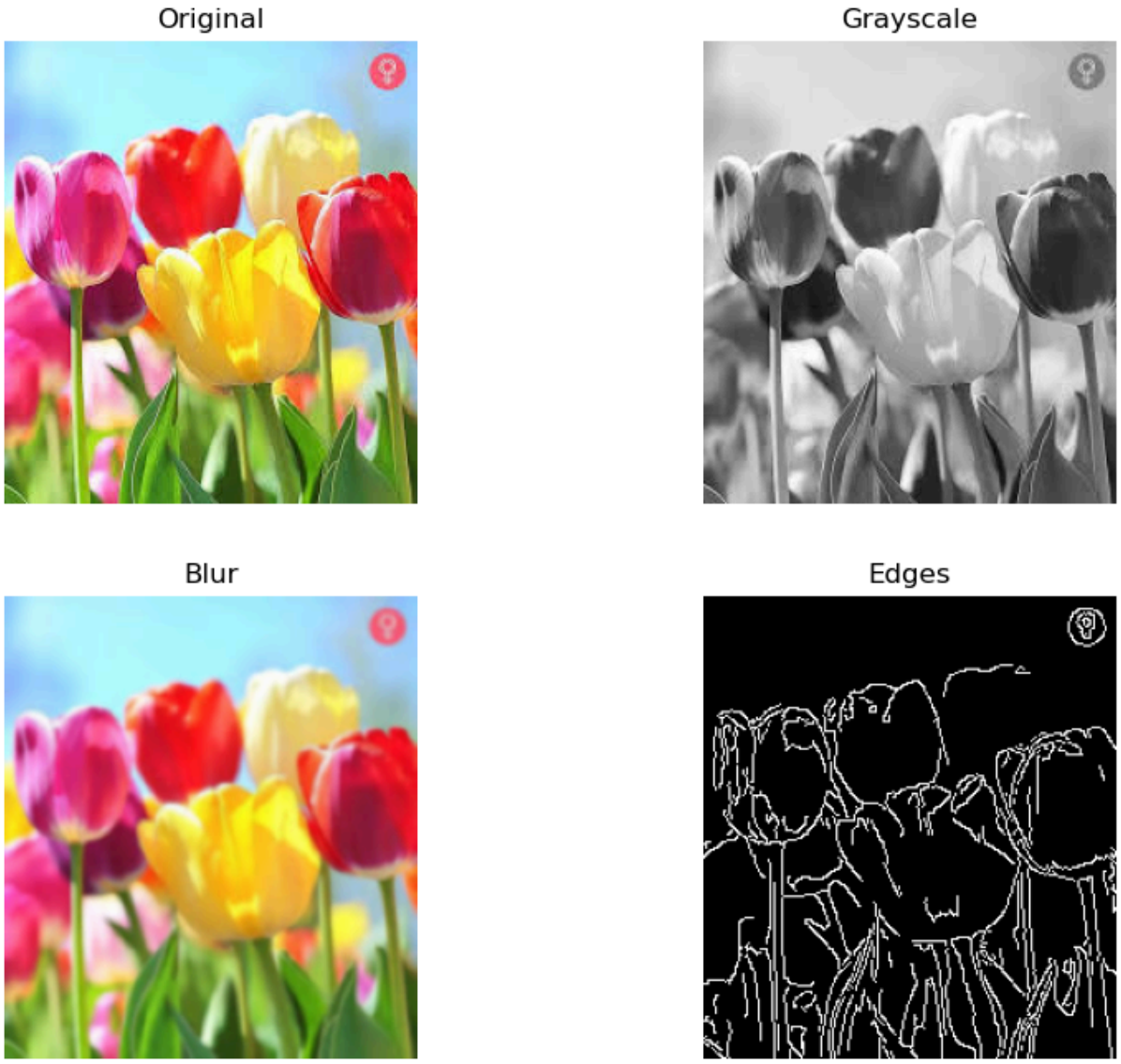
In [29]:
```
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [30]:
```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

In [31]:
```
blur = cv2.GaussianBlur(img, (5,5), 0)
blur_rgb = cv2.cvtColor(blur, cv2.COLOR_BGR2RGB)
```

In [32]:
```
edges = cv2.Canny(gray, 100, 200)
```

In [33]:
```
plt.figure(figsize=(10,8))

# Original
plt.subplot(2,2,1)
plt.imshow(img_rgb)
plt.title("Original")
plt.axis('off')

# Grayscale
plt.subplot(2,2,2)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale")
plt.axis('off')

# Blur
plt.subplot(2,2,3)
plt.imshow(blur_rgb)
plt.title("Blur")
plt.axis('off')

# Edges
plt.subplot(2,2,4)
plt.imshow(edges, cmap='gray')
plt.title("Edges")
plt.axis('off')

plt.show()
```



In [ ]: